



COURSE CONTENT

- 3.1.1 Introduction**
- 3.1.2 Thermal Jet Printer**
- 3.1.3 Sanders Model Maker**
- 3.1.4 Objet Quadra System**

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3.1.1 INTRODUCTION

Concept Modelling technologies represent a class of AM processes primarily developed for rapid visualization, design verification, and functional prototyping. Unlike industrial-grade AM processes that focus on producing high-strength, fully dense parts, concept modellers emphasize speed, ease of use, surface finish, and visual fidelity of printed prototypes. These systems are extensively used in product design studios, research environments, and industries requiring quick turnaround times for concept evaluation and aesthetic assessment.

Concept modelling systems generally employ liquid photopolymers (or) thermoplastic materials, which are selectively deposited or solidified layer by layer to form three-dimensional models directly from digital CAD data. They offer high-resolution outputs, smooth surface finishes, and the capability to produce intricate geometries without complex post-processing requirements.

In AM, 3D printers are broadly categorized into Functional Modellers and Concept Modellers based on their purpose, precision, and application. Functional Modellers, often referred to as industrial-grade 3D printers, are designed to produce durable, high-performance components suitable for functional testing or direct end-use. In contrast, Concept Modellers are low-cost, quick-prototyping printers used primarily for visual or form-and-fit evaluations during the early stages of product design.

The following table shows the difference between Functional and Concept Modellers in terms of key characteristics and applications.

Difference between Functional and Concept Modellers in terms of key characteristics and applications

<i>Parameter</i>	Functional Modellers (Industrial-grade 3D Printers)	Concept Modellers (Low-cost, Quick Prototyping Printers)
<i>Purpose</i>	Used to produce fully functional prototypes and end-use parts	Used for visual and conceptual models to demonstrate design intent
<i>Accuracy</i>	Very high dimensional accuracy and surface finish	Moderate accuracy, suitable for design validation only
<i>Material Range</i>	Wide range includes engineering-grade polymers, composites, and metals	Limited to basic thermoplastics like PLA, ABS, or resin
<i>Build Volume</i>	Large build envelopes for industrial-scale components	Smaller build sizes, ideal for desktop applications
<i>Strength of Parts</i>	Parts exhibit high mechanical strength and thermal stability	Parts are weaker, mainly suitable for form and fit testing
<i>Cost</i>	High initial investment and operational cost	Low-cost and affordable for educational or design offices
<i>Speed</i>	Relatively slower due to precision and multi-step processes	Faster printing suitable for quick visualization
<i>Applications</i>	Aerospace, automotive, tooling, and functional testing	Product design visualization, education, and concept presentation

The present chapter will focus on the following concept modelling technologies:

- Thermal Jet Printer: Uses thermal inkjet printing technology to deposit droplets of thermoplastic or wax-based materials layer by layer. The material solidifies upon cooling, enabling the creation of detailed prototypes with smooth surfaces.
- Sanders ModelMaker 3D Printer: Employs precision inkjet heads to deposit build and support waxes alternately, achieving very fine resolution and dimensional accuracy. The support wax is later melted away to reveal the finished part, making it ideal for high-detail concept models and patterns for casting.
- Objet Quadra Systems: Utilize PolyJet technology, where multiple nozzles jet layers of liquid photopolymer that are instantly cured using ultraviolet (UV) light. This enables multi-material and multi-color printing with excellent surface finish and high dimensional accuracy.

3.1.2 THERMAL JET PRINTER

The Thermal Inkjet Printer, also known as a ThermoJet Printer, operates on the principle of Material Jetting (MJ) technology. In this additive manufacturing process, the system uses inkjet-style printheads to deposit microscopic droplets of molten thermoplastic or wax material onto a build platform, where each droplet rapidly cools and solidifies to form a layer. Simultaneously, a support material, typically a soluble wax is deposited to sustain overhangs and complex geometries during fabrication. This layer-by-layer deposition continues until the complete model is formed, producing parts with excellent surface finish and fine detail. However, the resulting components are primarily suitable for design verification or casting pattern applications, as the materials lack the mechanical strength required for functional use.

The Thermal Inkjet Printing approach was originally developed to create high-precision wax patterns for investment casting and Jewellery prototyping, where superior accuracy and surface smoothness are essential. The core technology, known as Thermal Phase Change Inkjet Printing, was pioneered by Solidscape, initially founded as Sanders Prototype, Inc. in 1993 by Royden C. Sanders in New Hampshire, USA. The company's mission was to design high-resolution wax printers for rapid prototyping and to produce master moulds for critical casting applications in the jewellery, dental, and precision engineering industries. Over time, the company's ownership evolved: in 2011, it was acquired by Stratasys Ltd., a global leader in polymer-based additive manufacturing; and in 2018, Stratasys sold Solidscape to the Prodways Group of France, enabling the brand to refine its specialization in industrial wax-based 3D printing solutions.

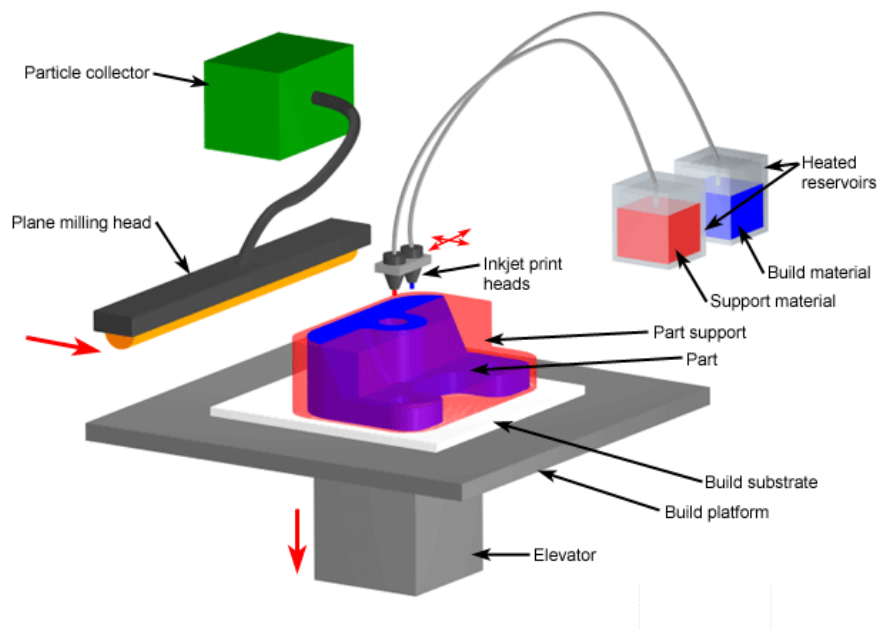
Thermal Inkjet Printing originates from 2D inkjet printer technology, where tiny droplets of liquid are ejected through nozzles to form images on paper. In 3D printing adaptation:

- The ink is replaced with molten thermoplastic or wax.
- The droplets are deposited layer by layer to form a 3D object.
- Upon deposition, the droplets instantly cool and solidify, forming a solid layer.

Because the material changes phase from liquid to solid during deposition, this process is often called "Thermal Inkjet Printing."

HARDWARE

- Dual Printheads: One for build (model) wax and another for support wax. Both operate independently and eject molten material droplets in the X–Y plane.
- Heated Material Reservoirs (Cartridges): Maintain wax materials at precise temperatures to ensure correct viscosity for jetting. Separate reservoirs for model and support materials.
- Build Platform (XY Stage): A precision-controlled platform that moves horizontally to position droplets accurately during deposition.
- Elevator (Z-axis Mechanism): Lowers the platform after each completed layer to maintain consistent layer thickness.
- Plane Milling Head: A micro-milling cutter smooths each deposited layer, ensuring uniform thickness and high-quality surface finish.
- Particle Collector (Vacuum System): Removes wax debris generated by the milling process and stores it in a collection bin.



Schematic of Thermal Ink jet printer

WORKING

- The Thermal Inkjet Printing process, employed by SolidScape, utilizes both a model wax and a support wax, which are first melted in separate reservoirs for material preparation.
- The core of the build process is Droplet Deposition, where high-precision printheads traverse the build platform, accurately ejecting microscopic droplets of the molten wax materials at specific coordinates to define the part's current layer cross-section.
- These droplets immediately undergo Solidification as they rapidly cool upon contact with the previous layer or the platform, forming a stable structure.
- A milling head precisely trims the surface of the newly deposited layer to ensure exceptional flatness and a uniform height before the next layer is added. Concurrently, a Particle Removal system, typically a vacuum, collects and filters the fine wax dust generated by the milling process.

- Following these steps, the build platform executes an incremental Layer Advancement (Z-axis motion), preparing for the repetition of the entire process until the model is complete.
- Finally, post-processing involves easily melting away the distinct support wax, resulting in a pristine, high-resolution wax pattern that is immediately ready for use in investment casting.

ADVANTAGES

- Exceptional accuracy up to ± 0.025 mm.
- Smooth surface finish suitable for direct mould making.
- Ideal for intricate designs with fine details and internal cavities.
- Clean burnout of wax patterns leaves no ash or residue.

LIMITATIONS

- Slow build speed compared to photopolymer or powder-based systems.
- Limited to wax-like materials, restricting mechanical applications.
- Fragile printed parts not suitable for functional testing.
- High maintenance cost due to precise temperature control and milling systems.

APPLICATIONS

- Jewellery Manufacturing: Creation of highly detailed wax models for lost-wax casting. Enables intricate, customized designs with fine surface fidelity.
- Investment Casting: Production of clean-burning wax masters for precision casting of metals and alloys. Ideal for small, detailed components.
- Dental Applications: Fabrication of wax patterns for crowns, bridges, and dental restorations where dimensional precision is critical.
- Industrial Prototyping: Production of form-and-fit models for evaluating complex geometries before final tooling.

3.1.3 SANDERS MODELMAKER

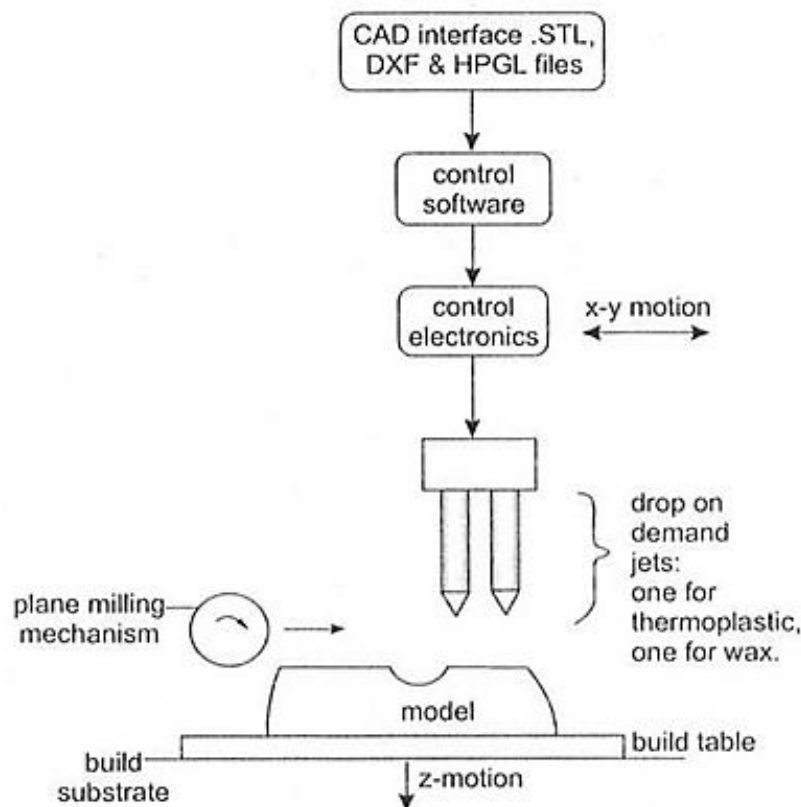
The Sanders Model Maker (MM) series represents one of the earliest and most precise implementations of inkjet-based additive manufacturing technology. Developed in the early 1990s by Sanders Prototype Inc., a subsidiary of Sanders Design in Wilton, New Hampshire, USA the Model Maker system was conceived to redefine accuracy and surface quality in rapid prototyping. Operating on the principle of layer-by-layer material jetting, the MM system deposits minute droplets of molten wax material to build complex three-dimensional geometries. Uniquely, the printer employs two separate inkjet heads one dedicated to the build material (model wax) and the other to a support material (sacrificial wax) both jetted concurrently to ensure dimensional stability and precise detailing, even in intricate or overhanging regions.

PRINCIPLE

The process starts by importing the 3D CAD model, which is sliced into thin layers by the system software. The machine heats the model and support wax materials to a precise

temperature and jets them alternately through fine drop-on-demand nozzles onto the build platform.

- The model wax forms the actual part geometry, while the support wax fills overhangs, cavities, or delicate features.
- After each layer is deposited, the material is solidified by cooling, ensuring a firm base for the next layer.
- This additive layering continues until the full 3D model is completed.
- Finally, the support wax is melted or dissolved away, leaving behind a smooth and highly detailed prototype ready for inspection or casting applications.



Schematic of Sanders Model Maker

ADVANTAGES

- Exceptional dimensional accuracy suitable for tooling and casting patterns.
- Produces parts with fine surface quality, reducing post-processing needs.
- Dual Material System – Simplifies the removal of support structures.
- Ideal for Investment Casting Patterns – Wax material makes it perfect for direct casting applications.

LIMITATIONS

- Layer-by-layer jetting and cooling make the process time-consuming.
- Restricted to wax materials, which are not suitable for functional testing.
- Wax models are delicate and require careful handling.
- Limits the size of parts that can be produced in a single build.

3.1.4 OBJET QUADRA SYSTEMS

The Objet Quadra System was one of the pioneering commercial 3D printers to implement what is now known as PolyJet technology, a process that combines inkjet printing principles with Ultraviolet (UV) photopolymer curing. Developed by Objet Geometries Ltd. In the early 2000s, the Quadra system was designed to produce high-resolution prototypes directly from CAD data, with exceptional surface finish and geometric accuracy. This innovation marked a major step in additive manufacturing, offering faster, cleaner, and more detailed prototyping than most competing technologies of its time.

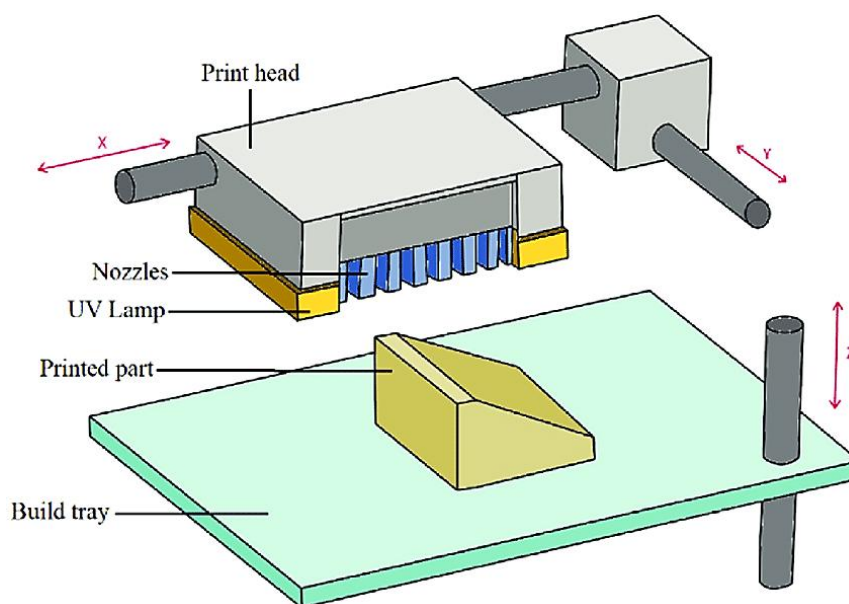
PRINCIPLE

The Quadra process operates on multi-nozzle inkjet printing, utilizing approximately 1,536 precision nozzles to jet ultra-thin layers of liquid photopolymer resin onto a build platform. Each layer is immediately cured by UV light, solidifying the resin before the next layer is deposited. This layer-by-layer curing process produces highly accurate, fully solid models without the need for post-curing or extensive post-processing.

Unlike traditional RP systems, the Quadra simultaneously jets two materials:

- Model material: A build resin forming the actual 3D geometry.
- Support material: A gel-like photopolymer that provides temporary support for overhangs and internal cavities.

Once printing is complete, the support material is manually or mechanically removed, leaving a smooth surface.



Schematic of Objet Quadra Systems

WORKING

The Objet Quadra process begins with importing the 3D CAD model, which is digitally sliced into thin layers to prepare it for printing. During fabrication, the system's printhead jets microscopic droplets of liquid photopolymer onto the build tray with exceptional precision. Simultaneously, integrated UV lamps instantly cure each deposited layer, solidifying the material as it is applied. This layer-by-layer fabrication continues until

the complete part is built. Once printing is complete, the support material is easily removed either manually or using water jets, revealing the final, high-precision model with a smooth surface finish and intricate details.

FEATURES

- High Resolution: Layer thickness as low as 16 microns (0.016 mm).
- Multi-Material Capability: Uses model and support materials simultaneously.
- Accurate and Smooth: Excellent dimensional accuracy and surface finish.
- Clean Process: Fully enclosed build chamber and automated operation.
- Multiple Jets: 1536 nozzles ensure fine detail and high build speed.

ADVANTAGES

- Exceptional accuracy and fine feature detail (± 0.1 mm typical).
- Smooth surfaces with minimal post-processing.
- Fast turnaround from digital model to physical prototype.
- Supports complex geometries, including internal cavities and thin walls.
- Efficient material handling through automated cartridge systems.

LIMITATIONS

- Restricted to photopolymer resins, which are less durable than engineering thermoplastics.
- Material cost and maintenance are relatively high.
- UV-cured materials may degrade or yellow under prolonged sunlight exposure.
- Limited mechanical strength and thermal resistance, restricting end-use applications.



COURSE CONTENT

- 3.2.1 Introduction**
- 3.2.2 Classification of Rapid Tooling**
- 3.2.3 Indirect Tooling Methods**
- 3.2.4 Direct Tooling Methods**

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3.2.1 INTRODUCTION

Rapid Tooling refers to the process of quickly designing and manufacturing tools, moulds, patterns (or) fixtures using AM technologies. It serves as a bridge between Rapid Prototyping and Conventional Manufacturing, facilitating faster product development and significantly reducing lead time in tooling fabrication.

In traditional manufacturing, producing tooling such as injection moulds (or) casting dies can take several weeks or even months and is often expensive due to the use of hardened steel and precision machining. During the product development phase, when design modifications are frequent, this traditional approach leads to delays and high costs.

Rapid Tooling addresses these limitations by utilizing additive manufacturing technologies to either:

- Directly produce the tooling components, or
- Create master patterns that are subsequently used in conventional moulding or casting processes.

This enables faster turnaround, lower cost, and greater design flexibility, making it especially useful for prototyping, small-batch production, and functional testing.

3.2.2 CLASSIFICATION OF RAPID TOOLING

1. Indirect Rapid Tooling:

- In Indirect Rapid Tooling, additive manufacturing is used to create a master pattern or model, which is then used to fabricate the actual tool using conventional processes like casting, moulding (or) electroforming.
- The pattern is first produced using AM technologies such as Stereolithography (SLA), Fused Deposition Modeling (FDM) (or) PolyJet printing.
- This pattern is then used to make a negative mould cavity by pouring materials like silicone rubber, epoxy resin, or metal-filled composites.

2. Direct Rapid Tooling:

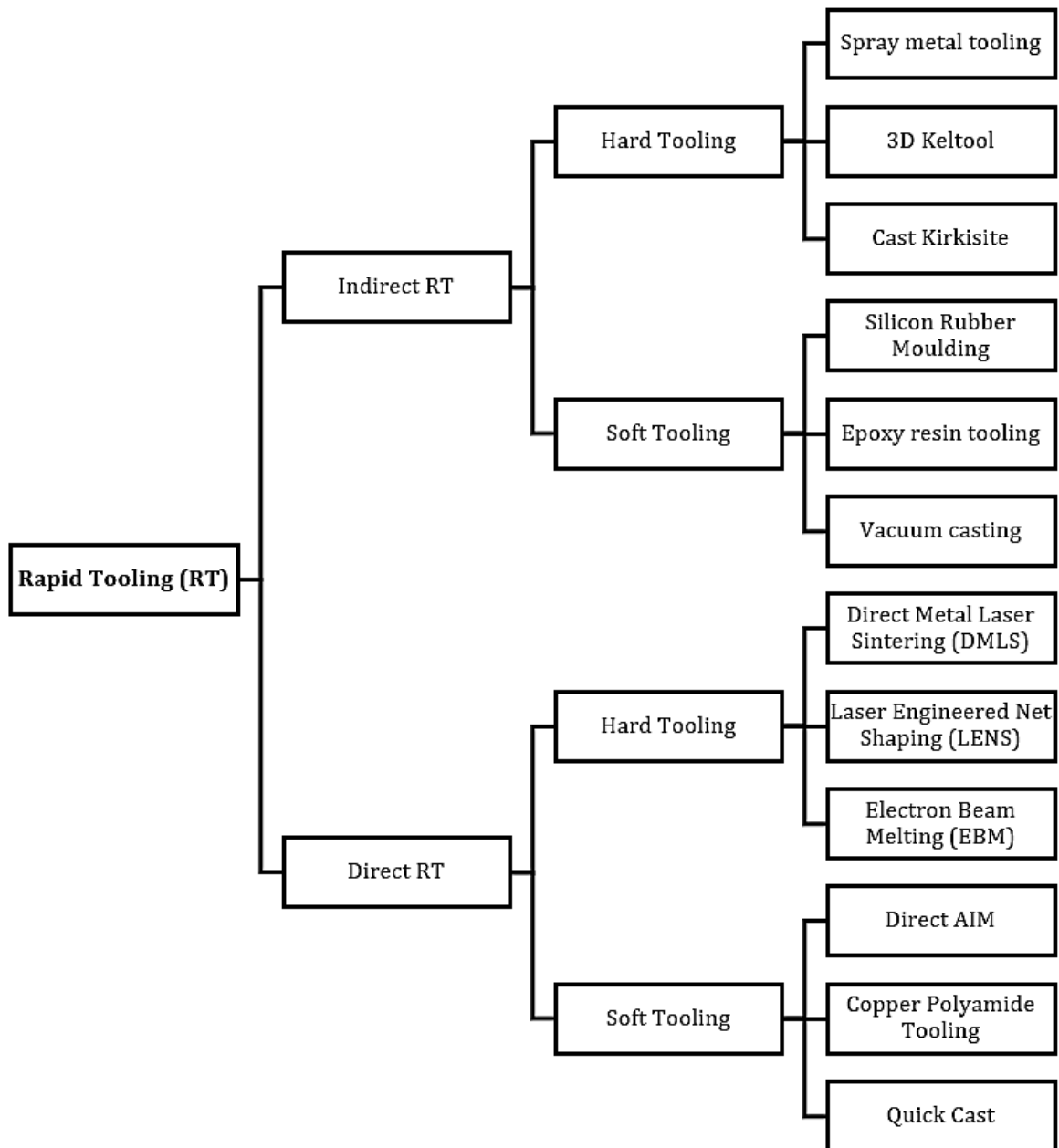
- In Direct Rapid Tooling, the tool or mould cavity is built directly using additive manufacturing technology, without the need for any intermediate pattern or model.
- The 3D CAD data of the mould (or) die is directly converted into a physical component using an AM process such as Selective Laser Melting (SLM), Direct Metal Laser Sintering (DMLS), (or) Electron Beam Melting (EBM).
- After printing, the tool may undergo post-processing steps such as machining, polishing, or heat treatment to improve surface finish and mechanical properties.

HARD TOOLING v/s SOFT TOOLING

Difference between Hard and Soft Tooling

<i>Aspect</i>	Hard Tooling	Soft Tooling
<i>Definition</i>	Tools produced using Additive Manufacturing made from metal to withstand high-volume production	Tools created rapidly using wax, polymer and resin for prototype or limited runs
<i>AM Processes used</i>	DMLS, SLM, or DED Processes	SLA, FDM, PolyJet processes
<i>Material Used</i>	Tool steels & Aluminum alloys	Wax, Epoxy, Silicone, photopolymer resins, or

		thermoplastics
Tool Life	Longer — can be used for hundreds to thousands of parts	Shorter — suitable for tens to a few hundred parts
Production Cost & Time	Higher cost and build time	Lower cost and very quick
Applications	Injection mould inserts, die-casting tools and forming dies	Prototype moulds, casting patterns, jigs and fixtures



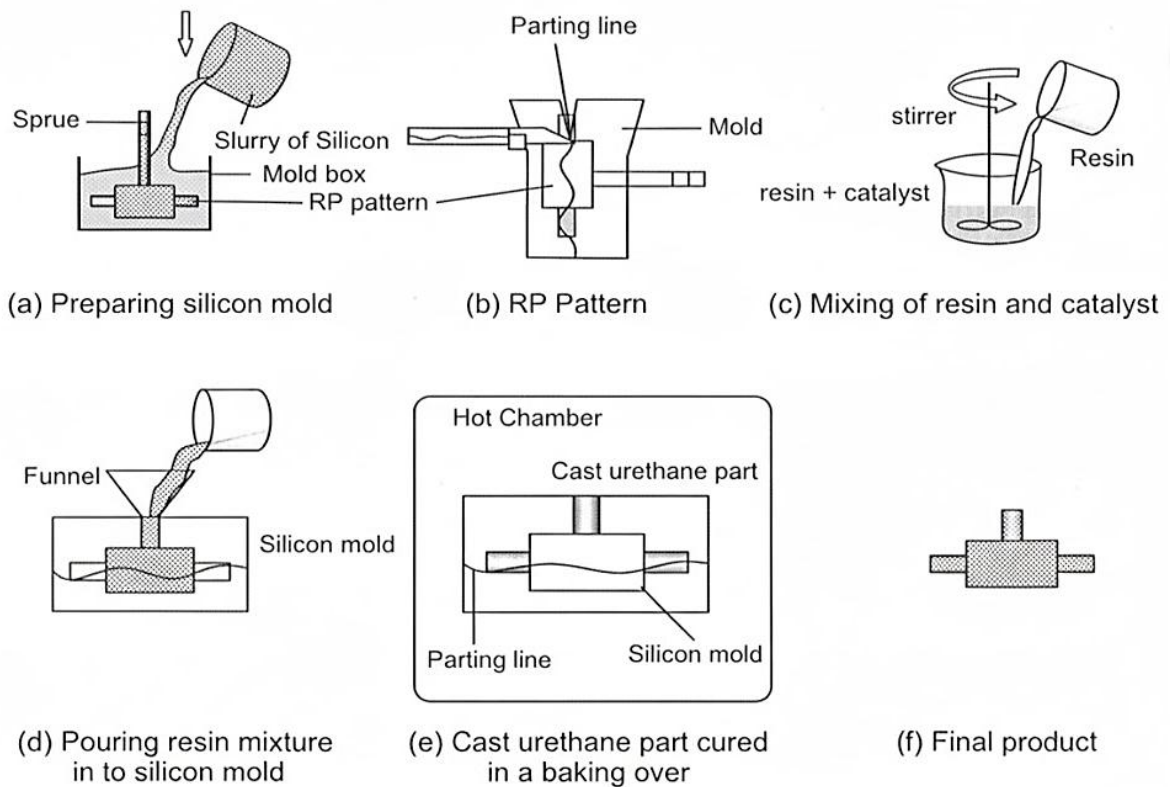
Classification of Rapid Tooling Methods

3.2.3 INDIRECT TOOLING METHODS

1. SILICONE RUBBER TOOLING:

- One of the most popular tooling applications for AM is the production of Room Temperature Vulcanizing (RTV) silicon rubber tooling.
- Involves creating flexible moulds using silicone rubber around a master model. An AM process is used to fabricate the pattern.

- The pattern is fixed into a holding box and coated with a special release agent (wax-based aerosol or petroleum jelly mixture) to prevent it from sticking to the silicon rubber mould.
- The Silicon rubber is then blended, vacuumed to remove air bubbles & poured into the box around the pattern until the pattern is completely encapsulated.



Steps in Silicone Rubber Tooling

- A master pattern is first created using an AM process and then placed inside a mould box. A slurry of liquid silicon rubber is poured around the pattern, allowing the silicon to flow and completely envelop it, filling all voids and surface details.
- After the silicon mould has cured, it is carefully cut open along the parting line, and the RP master pattern is removed. This leaves behind a cavity in the silicon mould that is an exact negative replica of the RP pattern. The two halves of the mould are then reassembled and prepared for casting.
- A resin material, typically polyurethane (or) epoxy, is mixed with a catalyst to initiate the curing process. The mixture is thoroughly stirred to ensure uniformity and proper chemical reaction, which is essential for achieving consistent mechanical properties in the final part.
- The mixed resin and catalyst solution is then poured into the prepared silicone mould through a funnel.
- After pouring, the filled mould is placed in a hot chamber or baking oven where the resin is cured. The heat accelerates the chemical reaction between the resin and catalyst, solidifying the material to form the cast urethane part.
- Once curing is complete, the mould is opened, and the cast part is removed.

Advantages:

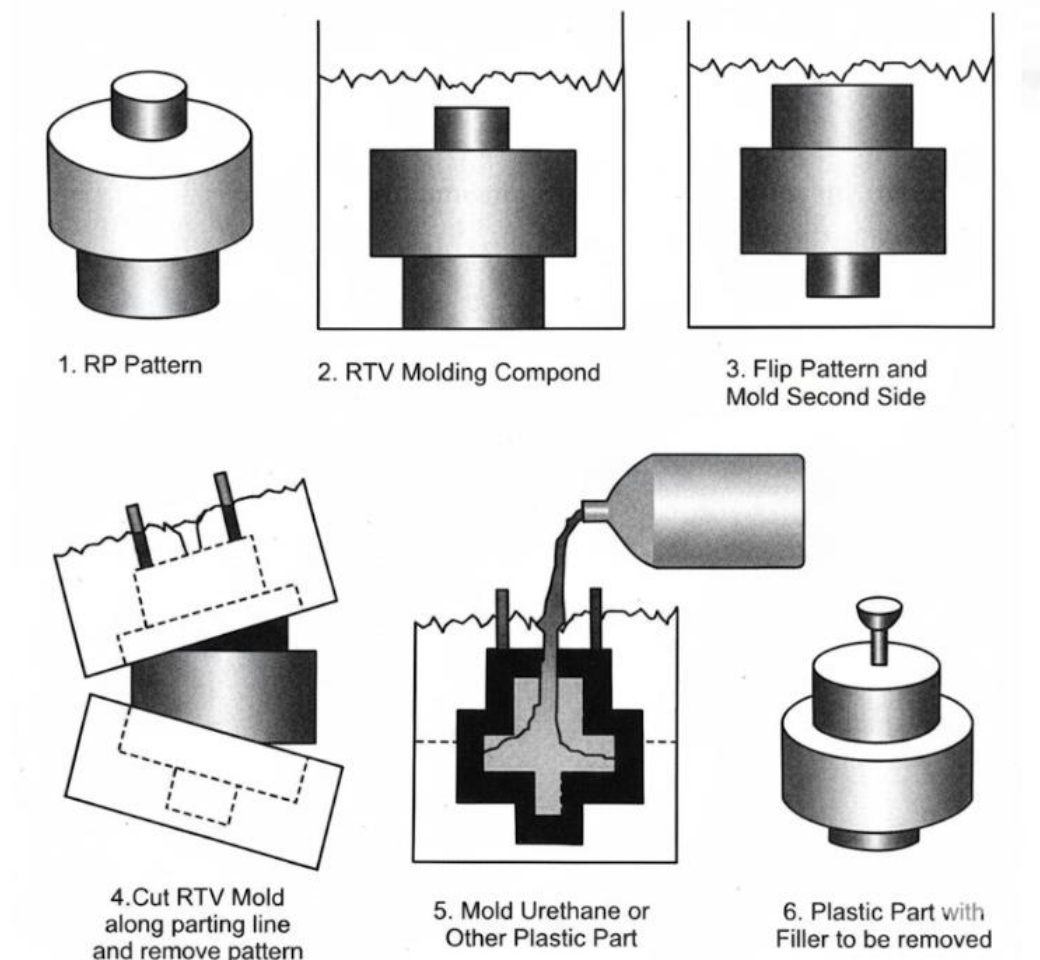
- Low cost and short lead time
- Excellent surface finish reproduction
- Flexible moulds for complex geometries

Limitations:

- Short tool life (~20–30 parts)
- Low thermal conductivity
- Not suitable for mass production

2. ALUMINIUM FILLED EPOXY TOOLING:

- It is also known as Composite Tooling.
- Aluminium-Filled Epoxy Tooling is a type of Room Temperature Vulcanization (RTV) process used for producing prototypes and short-run production moulds.
- It is a composite mould-making technique where epoxy resin acts as the matrix (binder) and aluminium powder serves as the filler material.
- Used to produce low-cost, moderately durable moulds or dies used in prototyping and small-batch production.
- The addition of aluminium improves the mechanical strength, thermal conductivity, and wear resistance of the epoxy.



Steps in Aluminium Filled Epoxy Tooling

Process:

- A Rapid Prototyped (RP) pattern typically made using AM process. This pattern represents the exact geometry of the part to be moulded. Its surface is cleaned and coated with a release agent to prevent sticking during epoxy casting.
- The master pattern is placed inside a mould box. A mixture of epoxy resin, aluminium filler & hardener is poured over the pattern to create the first half of the tooling. Because it cures at room temperature, it falls under the RTV category.
- After the first half of the epoxy tool cures, the mould is flipped. A release agent is applied to the cured first-half surface to prevent bonding. The aluminium-filled epoxy mixture is then poured to form the second half of the tooling.
- Once fully cured, the two halves of the aluminium-filled epoxy mould are separated carefully. The RP master pattern is removed, leaving behind a high-detail cavity that will be used to mould prototype parts.
- The cavity created inside the aluminium-filled epoxy mould is then used for casting prototype parts. Typically, urethane resins or low-temperature plastics are poured into the cavity. The higher thermal conductivity of the aluminium filler allows better heat dissipation, faster curing, and improved dimensional stability.
- Once the cast resin cures, the mould halves are opened, and the final plastic part is removed. This tool can be reused multiple times for short-run or prototype production.

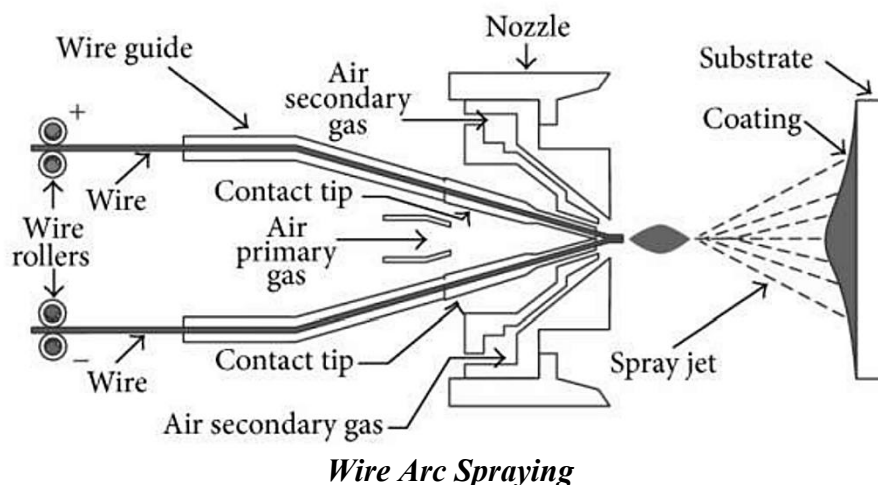
Advantages:

- Economical compared to metal tooling; suitable for prototype or short-run applications.
- Aluminium filler enhances heat transfer.
- Easier to work with than metal molds.
- Can be easily drilled, cut, or repaired if damaged.

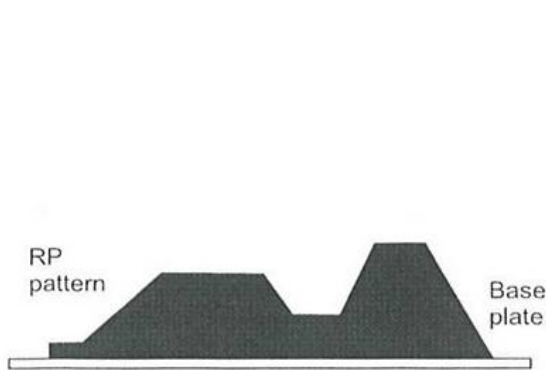
Limitations:

- Not suitable for high-volume or high-pressure production.
- Prolonged exposure to elevated temperatures may degrade epoxy.
- Minor dimensional changes may occur during curing.

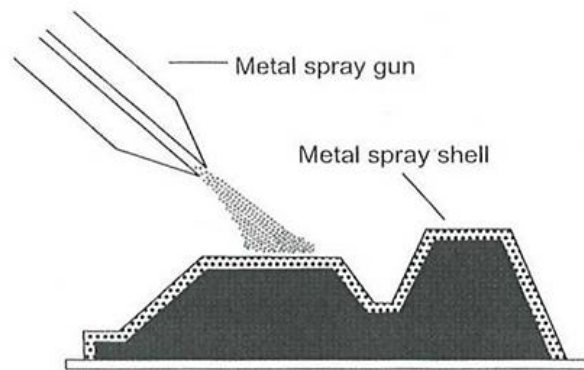
3. SPRAY METAL TOOLING:



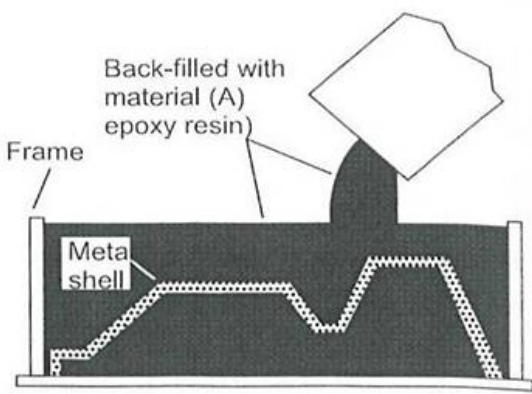
- Spray metal tooling involves depositing a thin layer of molten metal such as aluminium, zinc, (or) steel alloys onto a master pattern using a thermal spray gun.
- Also called Metal Spraying or Spray Deposition Tooling, used to produce metal-faced moulds or dies.
- This process provides a durable metal surface combined with low-cost backing material.
- Once the metal coating solidifies, it forms a hard metallic shell that accurately replicates the pattern's surface.
- This shell is then backed with resin, plaster, or epoxy to give the tool structural strength.



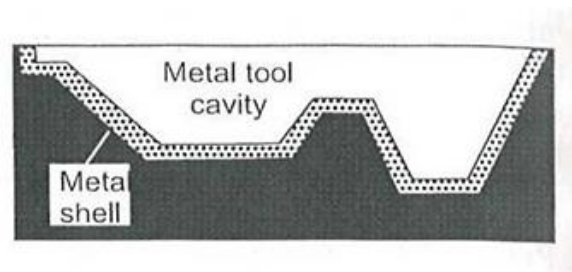
(a) Master pattern prepared by AM Process



(b) Metal Spraying



(c) Backing with epoxy resin



(d) Mould cavity ready after curing

Steps in Spray Metal Tooling

Process:

- The master pattern (usually made of wax, plastic, or epoxy) is fabricated by AM process, cleaned and coated with a release agent.
- Molten Aluminium (or) zinc is sprayed using a thermal spray gun. The metal droplets impact and solidify rapidly on the surface, forming a metallic shell 1-3 mm thick.
- The thin metal shell is backed with epoxy, aluminium-filled resin (or) metal-filled polyurethane for mechanical support.
- After backing material solidifies, the tool is separated from the master. The surface is then polished, machined, or drilled as required.

Advantages:

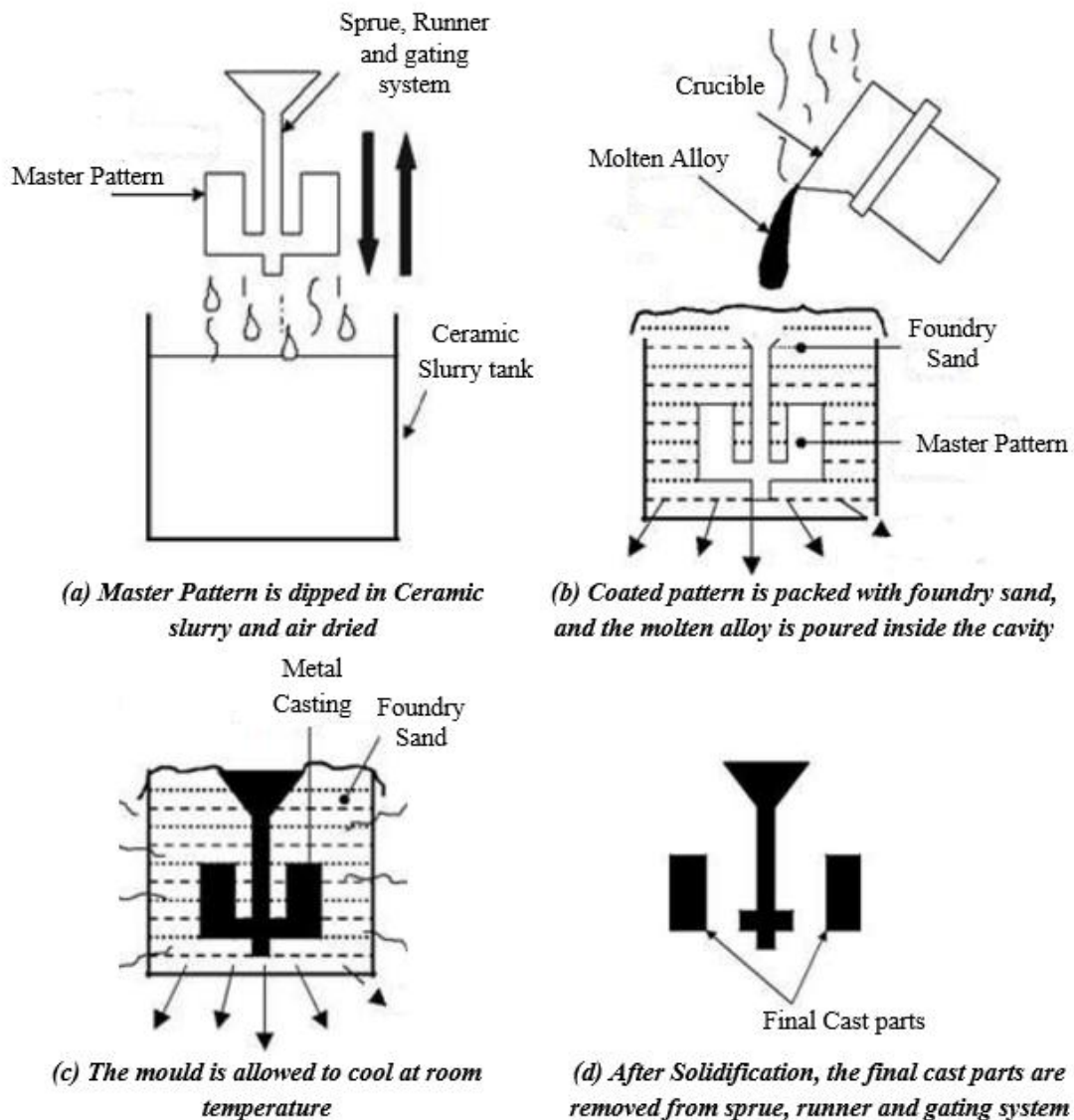
- Provides excellent wear resistance and durability.
- Longer tool life compared to epoxy tools.
- Faster and cheaper than machining full metal molds.
- Damaged surfaces can be re-sprayed or patched.

Limitations:

- Requires specialized spray equipment and expertise.
- Thin metal coating may fail under extreme load or temperature.
- Poor adhesion between metal and backing material can lead to delamination.

4. CAST KIRKISITE:

- Cast Kirkisite is a low-melting-point zinc-based alloy commonly used for prototype tooling, moulds and dies.
- It offers excellent dimensional accuracy, good surface finish, and can be easily cast or machined.



Steps in Cast Kirkisite

- Typical composition includes zinc, aluminum, and copper, and it melts around 385–400°C.
- Kirkisite is a zinc–aluminum–copper alloy with trace amounts of magnesium. Its typical composition consists of Zinc (Zn): 85–90%, Aluminum (Al): 4–8%, Copper (Cu): 1–3%, and Magnesium (Mg): less than 1%. After its service life, Kirkisite can be remelted and recast, making it an economical tooling material.

Process:

- A master pattern of the required component is created using AM process. The surface of the pattern is smoothed and coated with a parting agent to prevent sticking during the casting process.
- The prepared pattern is placed inside a moulding box, and foundry sand is packed around it to form the mould cavity. The mould is then dried to achieve the necessary strength for casting.
- Kirkisite ingots are melted in a crucible at a temperature of about 385–400°C. The molten alloy is carefully poured into the mould cavity surrounding the master pattern. Due to its low melting temperature, the alloy does not damage or distort the pattern material.
- The filled mould is allowed to cool at room temperature until the alloy solidifies completely. Controlled cooling helps minimize shrinkage, distortion, and internal stresses.
- Once solidified, the mould material is carefully broken away, and the cast Kirkisite tool is removed. The tool's surface may require light machining and finishing to achieve the desired surface quality.
- The completed Kirkisite tool is then ready to be mounted in moulding or forming machines for prototype production or short-run manufacturing.

5. 3D KELTOOL:

- 3D Keltool is an Indirect RT process developed to produce metal moulds from a Stereolithography (SLA) master pattern.
- Presently owned & licensed by 3D system.
- The word "Keltool" refers to the proprietary powder metal sintering process, which involves infiltrating a fused metal part with copper alloy.
- This alloy fills in the voids in the otherwise porous material, producing a surface with the finish and hardness necessary for an injection mould.
- The 3D Keltool process produces steel inserts or moulds suitable for Injection moulding, Die casting and Thermoforming.

Process:

- A high-accuracy SLA master pattern of the desired mould cavity is created by AM process.
- The SLA master is used to form a flexible silicone mould.
- A mixture of steel powder and epoxy resin is poured into the silicone mould and cured to form a green tool.
- The epoxy binder is burned out in a furnace (~800-900° C), leaving a porous steel structure.

- The porous structure is infiltrated with molten copper, resulting in a dense steel-copper composite insert (~60–70% steel, 30–40% copper).
- The infiltrated insert is finished, polished, and fitted into a mould base for production use.

Advantages:

- Produces metal moulds with high strength and thermal properties.
- Good surface finish and dimensional accuracy from SLA master.
- Lower cost and shorter lead time than CNC or EDM machined steel moulds.
- Repairable and machinable after infiltration.

Limitations:

- Relatively longer processing time (1–2 weeks).
- Shrinkage (~1–2%) during sintering and infiltration.
- Limited to moderate-size moulds (typically < 300 mm).

3.2.4 DIRECT TOOLING METHODS

1. DIRECT AIM:

- AIM stands for ACES Injection Moulding, where ACES denotes Accurate Clear Epoxy Solid.
- Direct AIM is a rapid tooling technique in which the mould cavity is fabricated directly using a Stereolithography (SLA) machine.
- This process enables the direct creation of moulds suitable for short-run production and prototype injection moulding applications.
- In the AIM process, the mould is built layer by layer using the SLA technique, forming a negative image of the desired part geometry.
- The completed mould is generally split into two halves to create the full mould cavity.
- These SLA-produced mould cavities can be used for injection moulding with a variety of materials such as thermoplastics, aluminum-filled epoxy, and ceramic-filled composites.

Advantages:

- Quick fabrication of tooling directly from CAD data.
- Use of production-grade thermoplastics, enabling functional prototype parts.
- Reduced lead time compared to traditional metal tooling.

Limitations:

- Low tool strength limits the number of parts that can be produced.
- Risk of tool failure due to heat and pressure during injection.
- Suitable primarily for short production runs and prototyping.

2. COPPER POLYAMIDE TOOLING:

- Copper Polyamide Rapid Tooling is an indirect rapid tooling technique where moulds are produced using a copper-filled polyamide composite material.
- This material combines the ease of additive manufacturing with the enhanced thermal and mechanical properties imparted by copper particles.

- The mould insert is fabricated using Selective Laser Sintering (SLS) or other powder-based additive manufacturing processes.
- The part is built layer by layer from the copper polyamide composite powder.
- Once fabricated, the mould insert may undergo surface finishing and polishing to improve part ejection and surface quality.
- These moulds (or) mould inserts are then used for injection moulding or vacuum casting applications.

Advantages:

- Improved thermal conductivity compared to unfilled polymers, allowing better cooling and shorter cycle times.
- Lightweight and easier to machine or modify than metal tooling.
- Reduced cost and lead time relative to conventional steel or aluminum moulds.

Limitations:

- Lower mechanical strength and limited life span compared to metal moulds.
- Not suitable for high-pressure or high-temperature production runs.
- Thermal conductivity, though higher than plastic, remains lower than that of metals.

3. QUICKCAST PROCESS:

- The QuickCast process is a rapid prototyping technique developed for producing investment casting patterns directly using Stereolithography (SLA).
- It enables the quick creation of accurate, lightweight, and burn-out-friendly patterns that replace traditional wax models in the investment casting process.
- The 3D part model is created and converted into a QuickCast build style, which produces a hollow internal structure with a honeycomb (hexagonal) lattice pattern to reduce mass.
- The SLA machine builds the pattern layer by layer using a photopolymer resin.
- The printed pattern is coated with ceramic slurry to form a ceramic shell mould through a conventional investment casting process.
- During burnout, the resin pattern vaporizes cleanly, leaving minimal ash and a precise negative cavity inside the ceramic shell.
- The molten metal is then poured into the shell to produce the final casting.

Advantages:

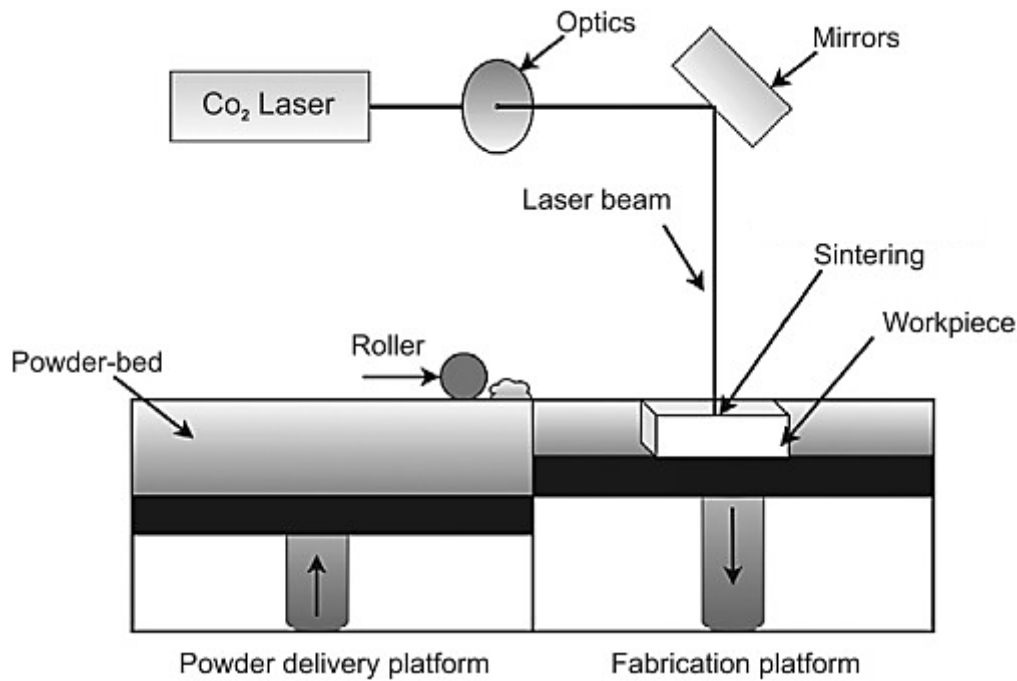
- High dimensional accuracy suitable for precision castings.
- Clean burnout, minimizing ash residue and defects in castings.
- Reduces lead time and cost compared to traditional wax tooling.

Limitations:

- High material cost compared to traditional wax patterns.
- Limited size depending on the SLA machine build volume.
- Handling care is required as SLA resins are brittle.

4. DIRECT METAL LASER SINTERING (DMLS):

- DMLS is an AM process that produces fully functional metal parts or tooling inserts directly from CAD data without the need for intermediate patterns or moulds.
- When applied to direct tooling, DMLS enables the creation of mould and die inserts made of metallic materials, suitable for injection moulding, die casting, and stamping applications.



Principle of Direct Metal Laser Sintering

- The mould or tool geometry is designed in CAD, incorporating features such as conformal cooling channels, texturing, and draft angles for improved functionality.
- A thin layer of metal powder (typically 20–40 μm thick) is spread across the build platform.
- A high-power laser selectively fuses (melts) the metal powder according to the digital cross-section of the tool insert.
- The platform then lowers, and the process repeats layer by layer until the complete tool is built.
- The part printed is subjected to heat treatment to improve mechanical properties.
- Machining, polishing, or finishing to achieve tight tolerances and surface finish suitable for moulding operations.

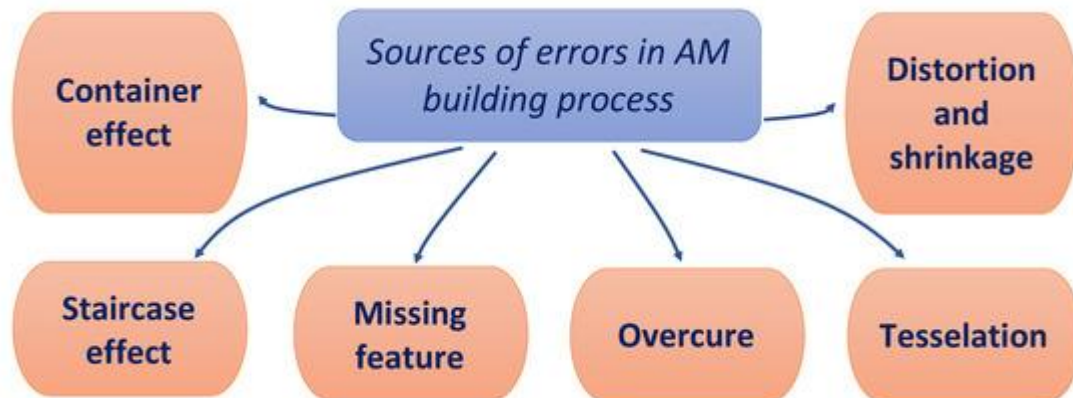
Advantages:

- Enables direct fabrication of metal tools and inserts without intermediate steps.
- Provides high dimensional accuracy and excellent detail reproduction.
- Produces dense metal parts with mechanical properties comparable to wrought materials.
- Offers high design flexibility for complex and customized tool geometries.

Limitations:

- High equipment and material costs make it expensive for tooling.
- Build size is limited by the printer's chamber capacity.
- Requires post-processing such as heat treatment, machining and polishing to achieve desired surface finish.
- Possibility of residual stresses and distortion due to rapid heating and cooling during laser sintering.

RAPID MANUFACTURING PROCESS OPTIMIZATION



COURSE CONTENT

- 3.3.1 Introduction**
- 3.3.2 Errors during Data Preparation**
- 3.3.3 Errors during the Part building**
- 3.3.4 Errors during Part finishing**

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3.3.1 INTRODUCTION

Accuracy in 3D printing refers to the degree to which the printed part's dimensions and geometry match the original CAD model. It is commonly measured in terms of dimensional deviation or tolerance, such as ± 0.1 mm or $\pm 0.2\%$ of the part size. Achieving high accuracy is essential for ensuring proper fit, function, and overall performance of additively manufactured components.

Optimizing the AM process plays a key role in improving dimensional accuracy, surface quality, and the mechanical performance of the final part. In AM, several factors influence overall process accuracy. These factors are generally classified into three major categories:

1. Errors during Data preparation: includes
 - STL file generation
 - Model slicing
 - Part build direction
2. Errors during the Part building:
 - Process-specific parameters such as layer thickness, scan speed, curing or melting behaviour and machine calibration etc.
3. Errors during Part finishing:
 - Surface finishing, cleaning, curing, and other secondary operations

Understanding and controlling these factors is crucial for producing parts that closely replicate the intended CAD design and meet the required functional specifications.

3.3.2 ERRORS DURING DATA PREPARATION

1. STL FILE ERRORS:

Errors caused by tessellation are often overlooked due to the common assumption that increasing the number of triangular facets will automatically minimize these errors. However, in practice, the number of triangles cannot be increased indefinitely due to limitations in file size, and processing speed.

Therefore, the resolution of STL files must be carefully controlled during their generation in a 3D CAD system by adjusting the tessellation parameters. During STL generation, two primary parameters influence accuracy:

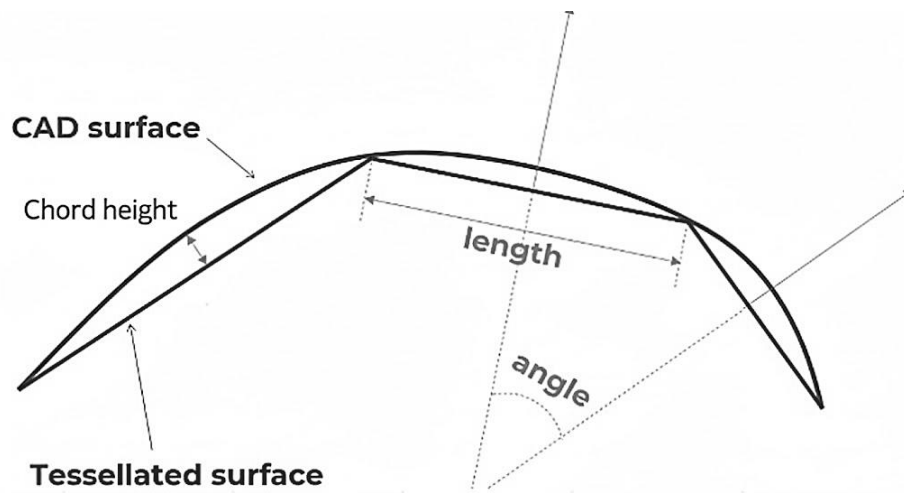
a. Chord Height:

- Chord height specifies the maximum allowable distance between the actual curved surface and the straight chord connecting two points on that surface.
- A smaller chord height reduces deviation from the actual geometry, resulting in higher accuracy.
- The upper limit of chord height depends on the overall model size, while the lower limit is governed by the accuracy of the CAD model itself.

b. Angle Control:

- Angle control defines the maximum permitted angle between the normals of adjacent triangular facets.
- A low angular tolerance results in high accuracy and smoother surfaces but produces a larger STL file size.

- In contrast, a high chord height combined with a high angular tolerance leads to lower accuracy and more faceted surfaces but generates a smaller file size.



Tessellation Error in CAD

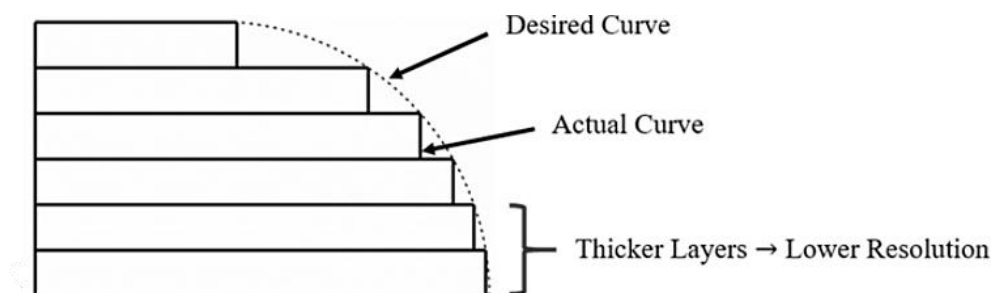
Reducing chord height and tightening angular tolerance improves geometric fidelity and produces smoother surfaces. However, this also increases the number of facets, resulting in significantly larger STL files and higher processing requirements. Therefore, it is essential to choose parameter values that balance accuracy with file size.

2. SLICING ERRORS:

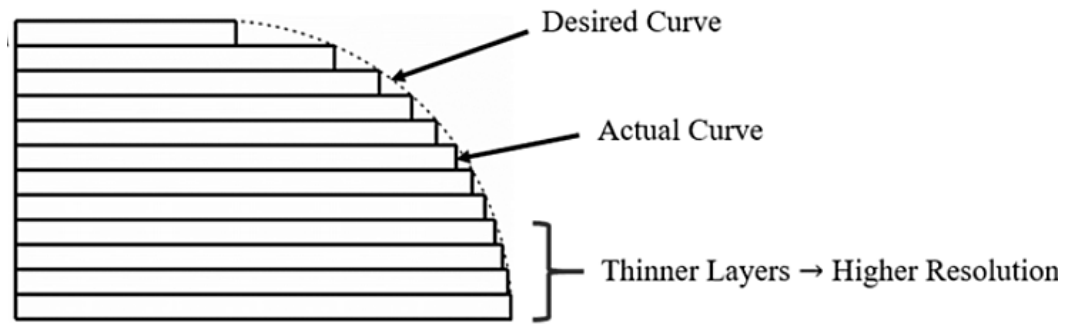
In AM, slicing is a critical step that significantly influences the dimensional and surface accuracy of the final part. AM systems fabricate components layer by layer, and this discrete layering inherently introduces the stair-stepping effect, particularly on curved or inclined surfaces. Because each layer has a finite thickness, the original CAD geometry cannot be reproduced continuously, instead it is approximated by step-like edges. This approximation leads to deviations between the desired curve and the actual printed contour, affecting overall part accuracy.

The magnitude of this error is strongly dependent on layer thickness:

- Thicker layers result in larger step heights, producing lower surface resolution and reduced dimensional accuracy.
- Thinner layers more closely follow the curvature of the CAD model, enhancing surface resolution and improving the accuracy of the printed part though at the cost of increased build time.



Thick layers - reduced dimensional accuracy

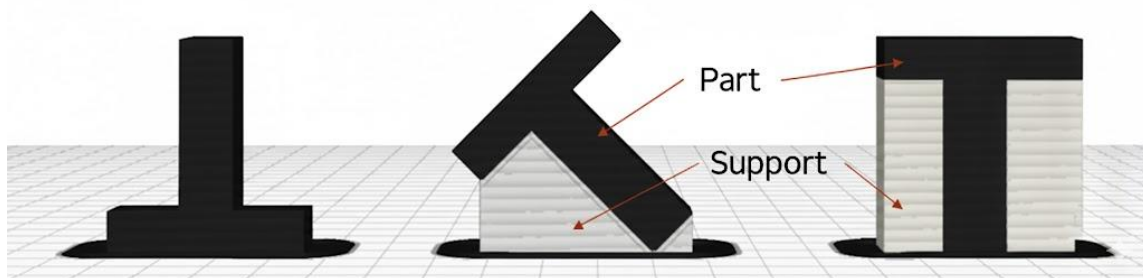


Thin layers - improved dimensional accuracy

Curved profiles in a CAD model are mathematically defined by their curvature and radius of curvature. For accuracy analysis, especially in Additive Manufacturing, these curves are often approximated using arc segments. Arcs provide simpler mathematical handling, constant curvature, easier error estimation, and better control over tessellation and slicing parameters. This makes them highly suitable for evaluating stair-stepping effects and understanding how closely the printed layers follow the intended geometry.

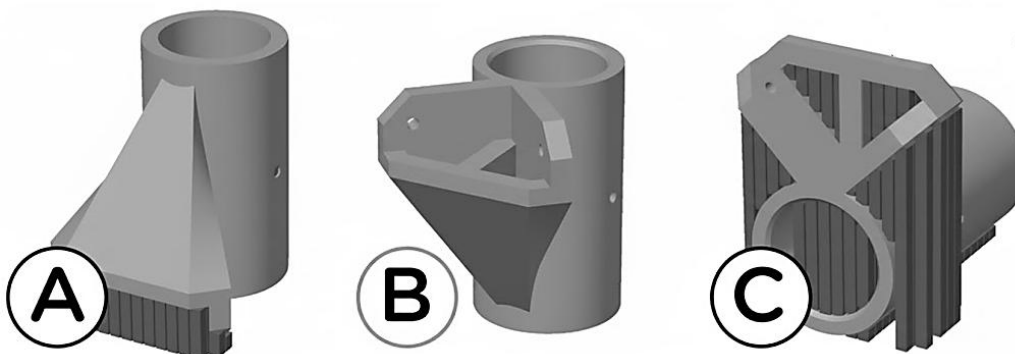
3. PART BUILDING ORIENTATION:

The selection of part build orientation is one of the most critical decisions in AM, as it directly influences accuracy, mechanical performance, build time, and overall manufacturing cost. Proper orientation helps minimize dimensional errors, reduce surface defects, and achieve better structural integrity. Since AM parts generally exhibit weaker mechanical properties along the Z-axis, orienting the part to place major loads within the XY plane significantly enhances strength.

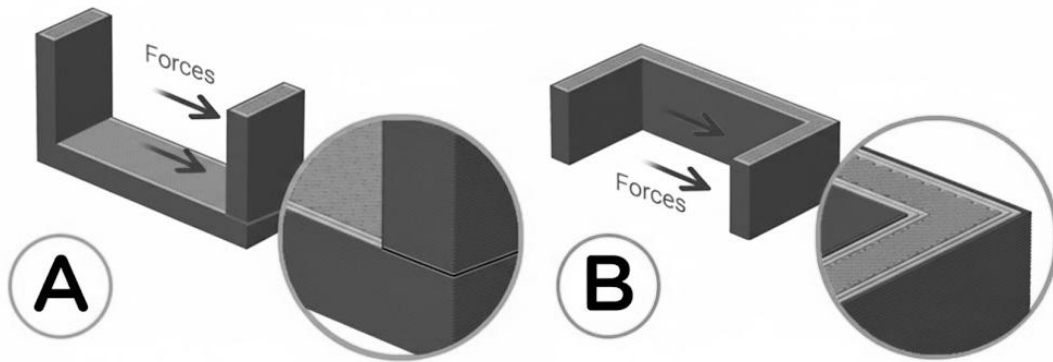


Part build orientation

Example 1: The part in Orientation B is ideal because it minimizes or eliminates the need for support structures, unlike orientations A and C, which require excessive supports.



Example 2: The part should also be printed in orientation B, where the applied forces act across the layers, resulting in stronger walls. Printing in orientation A makes the part weak because forces align with the layer lines, causing the walls to snap off easily.



An optimal build orientation should aim to:

- Minimize the height of support structures, reducing material usage and post-processing effort.
- Reduce the overall height of the part, thereby decreasing the number of layers and shortening build time.
- Limit inclined surfaces, which are more prone to stair-stepping errors and poor surface finish.
- Align critical load-bearing directions parallel to the layer plane (XY) to improve mechanical performance and durability.

Guidelines for Feature Orientation in AM:

- Holes are oriented orthogonally to the horizontal planes to prevent difficult support removal and to minimize stair-stepping on the internal surfaces.
- Cuts that pass entirely through the part or exceed a minimum depth are aligned such that the intersecting planes become horizontal; shallow or partial cuts are generally ignored to avoid unnecessary complexity.
- Shaft features are oriented so that their defining planes lie horizontally, ensuring improved external surface finish and reduced stair-stepping.
- Protrusions formed by revolving a profile are positioned with their axes vertically aligned to achieve better accuracy and reduce surface imperfections.
- Shell features are oriented with the concave surface facing upward to reduce the need for internal supports and improve the quality of inner surfaces.
- All axes within the model are aligned vertically wherever possible to enhance build stability, accuracy, and surface finish.

3.3.3 ERRORS DURING THE PART BUILDING

(a) PART BUILDING ERRORS IN SLA:

In SLA, part-building accuracy is influenced by errors occurring during the curing and scanning stages. These errors affect dimensional precision, surface finish, and mechanical strength. The major types of SLA part-building errors include:

1. Curing Errors: Curing errors occur due to improper laser exposure during the photopolymerization process.

- Overcuring: A controlled amount of overcuring is necessary to ensure proper bonding between layers and achieve fully solid parts. However, excessive laser exposure causes the resin to cure beyond the intended boundaries. It can lead to Dimensional inaccuracies, Positional errors and loss of fine details and sharp edges.
- Undercuring: Occurs when laser exposure is insufficient to fully cure the resin in each layer. The resin remains partially polymerized, reducing overall strength and layer adhesion. Undercuring can cause Weak or brittle parts, delamination between layers and distorted geometry

2. Control Errors: Control errors are associated with layer thickness selection and scan positioning accuracy.

- Layer Thickness: Using large layer thickness helps reduce build time, but it leads to noticeable stair-stepping on curved or inclined surfaces, poor surface finish that requires more post-processing and weaker interlayer bonding, which lowers mechanical strength. In contrast, using small layer thickness improves accuracy but increases the overall build time, makes the process more sensitive to exposure variations that may result in overcuring and risk of print failure.
- Scan Position Control: Accurate scan positioning ensures that each point is cured exactly at its designated location. Any deviation or misalignment introduces positional errors, geometric distortions, and dimensional deviation in the part. Poor scan control may also lead to stitching errors between scan paths.

(b) PART BUILDING ERRORS IN FDM:

1. Warping: Warping occurs when printed layers cool unevenly, causing thermal contraction that lifts the corners of the part from the build plate. It is common in high-shrink materials like ABS and Nylon and results in dimensional inaccuracies. This defect can be minimized by using a heated bed, enclosing the printer, adding brims or rafts and reducing fan cooling during the initial layers.

2. Layer Shift: Layer shift happens when an entire layer is printed in the wrong position due to unintended movement of the print head or build platform. It is usually caused by loose belts or pulleys, stepper motor skipping, excessive vibrations. The result is misaligned geometry and distorted features, which can be prevented by tightening mechanical components, reducing print speed and lubricating rails.

3. Poor Interlayer Bonding: Poor interlayer bonding occurs when successive layers do not fuse properly, producing weak prints. This defect is typically caused by insufficient nozzle temperature, excessive print speed and moisture-laden filament. It leads to cracks between layers and reduced strength along the Z-axis and can be prevented by using the correct print temperature, slowing down printing and storing filament in a dry environment.

4. Surface Roughness: Surface roughness can be caused by large layer height, inconsistent extrusion flow, worn nozzles, and improper print speed settings. This defect affects dimensional accuracy and aesthetics, but can be improved by selecting smaller layer heights, maintaining the extrusion system, and optimizing speed and temperature settings.

5. Nozzle Clogging: Nozzle clogging happens when the nozzle becomes partially or fully blocked, preventing smooth extrusion of filament. It is caused by dust, degraded filament, moisture in the filament and incorrect temperature settings. This leads to under-extrusion, missing layers, and possible print failure, and can be prevented by using clean and dry filament, maintaining proper temperature, and regularly cleaning or replacing the nozzle.

(c) PART BUILDING ERRORS IN SLS:

1. Powder Bed Defects: Powder bed defects occur when the powder is not evenly distributed across the build platform. Uneven powder spreading leads to variations in layer thickness, poor surface quality and porosity. Common causes include worn recoater blades, contaminated powder, moisture, and improper powder flowability.

2. Layer Irregularities: Layer irregularities arise when each deposited powder layer is not uniform in thickness. This leads to inconsistent energy absorption during laser scanning, causing distorted geometry, reduced dimensional accuracy, and weak zones within the part.

3. Thermal Distortion: Because SLS involves high temperatures, uneven heating and cooling can cause warping or curling of the part during building. Differences in thermal expansion can create internal stresses that distort the geometry or cause layer separation.

4. Incomplete Sintering: If the laser power or scan speed is not properly controlled, the powder may not fully fuse, leading to weak bonding between particles. This results in fragile parts, poor mechanical strength, and grainy or crumbly surfaces.

5. Over-sintering: Excessive laser exposure causes the powder to melt beyond the intended boundaries. This leads to dimensional inaccuracy, loss of fine features, surface roughness, and possible fuse-together of moving parts designed with clearance.

3.3.4 ERRORS DURING PART FINISHING

Part finishing is an essential stage in AM, especially for producing high-quality models and master patterns used in indirect tooling. Part finishing is a crucial phase in AM, particularly for components intended for high-quality visual models, functional prototypes, or master patterns used in indirect tooling. Finishing operations enhance both the surface appearance and dimensional accuracy of the printed part.

A variety of processes can be used to finish AM models. Common finishing methods include wet and dry sanding, sand blasting, coating, spray painting, infiltration with specialized solutions, machining, and several others.

Finishing techniques that offer better dimensional control tend to have less impact on the accuracy of the final model. For example, machining or milling-based finishing produces more precise results and causes minimal dimensional deviation, whereas manual wet sanding or sand blasting can remove more material unevenly, thereby having a greater influence on the model's overall accuracy.

Key sources of finishing-related errors include:

- Stair-Stepping Removal: Curved and sloped surfaces exhibit visible stair-stepping due to layer-by-layer fabrication. Removing these steps through sanding, polishing, bead blasting, or vapor smoothing is essential to achieve a smooth and accurate surface.

- Support Removal Damage: Poor support design or excessive support structures can result in surface marks and dents during removal.
- Dimensional variation: Techniques such as sanding or grinding can unintentionally remove excess material, leading to loss of dimensional accuracy. Over-polishing can distort edges, radii, and fine features.
- Thermal & Chemical finishing Issues: Chemical treatments may cause over-melting, softening edges or altering tolerances. Heat-based smoothing can lead to warping or uneven gloss levels.
- Incomplete Cleaning: Powder-based AM processes may leave embedded powder or un sintered particles, affecting surface quality and coating adhesion.